

Sensor Fusion in stbox-viz

FP-SNS-STBOX1 / SDDataLogFileX · SensorTile.box PRO · 28 April 2026

Input data we record per session

Every recorded session writes up to four CSV/WAV files to the SD card. The sensors and their roles:

Source	Rate	What it gives us	Used in fusion?
LSM6DSV16X — 3D gyroscope	100 Hz	Angular velocity X/Y/Z, ± 500 dps full scale, 17.5 mdps/LSB	Yes — Madgwick prediction step
LSM6DSV16X — 3D accelerometer	100 Hz	Specific force X/Y/Z, ± 4 g full scale, 0.122 mg/LSB	Yes — Madgwick correction step
LIS2MDL — 3D magnetometer (compass)	100 Hz	Magnetic field X/Y/Z (mgauss). Offset cancellation + temp comp + LPF on.	No — iron-distorted, see below
LPS22DF — barometer	100 Hz	Pressure (hPa) and temperature. Used for board height above water.	No — downstream height pipeline only
u-blox MAX-M10S — GPS position	10 Hz	Lat/lon/alt + UTC time, NMEA GGA + RMC	No — ride detection, map, height anchor
u-blox MAX-M10S — GPS speed	10 Hz	Position-derived (haversine), <i>not</i> module Doppler-speed (unreliable)	No — ride detection + push-off trigger

The fusion filter

We use a **6DOF Madgwick AHRS** filter — gyro + accel only. Implementation:

`Utilities/rust/stbox-viz/src/fusion.rs` (port of the classic Madgwick 2010 IMU variant).

Magnetometer is *deliberately excluded*; the other four signals are used *downstream* of fusion, not as inputs to it.

Why we drop the compass

Validation against GPS course-over-ground (Ermioni, 22.4.2026):

- median residual: **-14°**
- p5–p95 spread: **307°** — essentially random

The LIS2MDL is hard/soft-iron-distorted by ferromagnetic mounting hardware. Feeding it into a 9DOF gradient would couple heading garbage into roll & pitch and corrupt the whole quaternion. So we accept that *yaw has no absolute reference*. The fix is hardware (plastic screws, redesigned enclosure away from steel mounts).

The math, per sample

1. Predict the quaternion derivative from the gyroscope alone: $\dot{q} = \frac{1}{2} \cdot q \otimes \omega$.

2. Compute the Madgwick gradient of the accelerometer error (the rotated world gravity vector should match the measured body-frame accel).
3. Subtract $\beta \cdot \nabla F / \|\nabla F\|$ from the gyro-predicted derivative.
4. Integrate by dt , then renormalise the quaternion.

β is the tuning knob — how strongly we trust accel vs gyro. dt is auto-detected from the median tick delta in the CSV (1 tick = 10 ms $\Rightarrow$ 100 Hz $\Rightarrow$ $dt = 0.01$ s).

Input units into Madgwick

- Accel: **mg**. Direction-only after normalisation.
- Gyro: **mdps**. Converted to rad/s before integration.

What downstream code does with each signal

Quaternion $\rightarrow$ Nasenwinkel

`nose_angle_series_deg` projects the body-Y axis (board nose direction) into world Z and takes `asin`:

```
nose_z = 2 * (q_j * q_k - q_s * q_i)
angle° = asin(nose_z)
```

Drift correction: 1 s centred rolling median (suppresses correction spikes) followed by subtraction of a 60 s rolling-median baseline. This drives the Nasenwinkel trace and the pump-cadence STFT.

Quaternion $\rightarrow$ 3D foil rendering

Mast-mount (`animate --mount mast`): orientation straight from the Madgwick quaternion with `quat_strip_yaw` removing the drifting yaw. Captures the 90° tail-place transient just before push-off because Madgwick integrates the gyro continuously.

Deck-mount (`animate --mount deck`): bypasses Madgwick and uses **accel-only tilt** — 0.25 s rolling mean on accel, then `pitch = atan2(-ay,  $\sqrt{a_x^2 + a_z^2}$ )`, `roll = atan2(ax, az)`.

Reason: gyro-Z bias survives `quat_strip_yaw` once the mast is tilted, making the rendered foil spin around its own mast on clean rides.

Barometer $\rightarrow$ height above water

Two-stage correction (`baro.rs`):

1. **Temperature compensation.** The LPS22DF sits in a semi-sealed enclosure, so pressure couples to temperature via the ideal-gas law. A 10 °C swing produces ~15 hPa of fake altitude — far larger than the 0.08 hPa signal from the 80 cm mast.
2. **GPS-anchored water reference.** When GPS speed < 3 km/h the board is in the water; we linear-interpolate the TC'd pressure across flying segments and compute height via the local hypsometric approximation $8434 \cdot (1 - P/P_{ref})$.

Final guards: clamp to physical mast range [-0.1, 0.9] m, NaN samples where 10 ms pressure delta exceeds 1 hPa (sensor glitches). NaN-aware smoothing keeps neighbouring real samples valid.

GPS → ride detection, map, push-off

- **Ride detection.** Sustained GPS speed > 3 km/h for ≥ 10 s, merge gaps < 30 s, pad ± 3 s. More reliable than the pitch-oscillation detector for smooth flying where the board barely pitches.
- **Map trace.** One Plotly trace per ride, coloured by speed (Viridis 0–25 km/h), hover shows UTC + speed + nose angle + height-above-water.
- **Push-off trigger.** First sample where a 1 s rolling mean of *raw* position-derived speed exceeds 4 km/h sustainedly for 0.5 s. Drives the phase-band shading (Tragen / Rennen / Foilen) and the Push-off-Winkel flash.
- **Speed cleaning.** Two-stage filter: reject samples with $|\Delta v|/\Delta t > 15$ km/h/s (multipath spikes), then 5 s rolling median. Module Doppler-speed is unreliable on this hardware (median 0.12 km/h while position deltas show 10–30 km/h), so we always derive speed from position.

The pump-cadence trace is still driven by the Madgwick output even in deck-mount mode — only the 3D mesh path uses the accel-only attitude. That keeps the pump fundamental (~0.5 Hz) clean while giving us a stable visual rendering.

Carry phase — explicitly hidden

During the carry phase (no GPS speed) the 3D mesh is hidden completely — the panel shows “*Tragen — keine 3D-Daten*”. Without an absolute heading reference (no usable mag, GPS course unreliable below ~3 km/h) any rendered orientation during carry is a guess; we'd rather show no foil than a wrong one. The 3D rendering switches on at *water_set_t* (sustained GPS speed > 4 km/h) or after a 10.5 s fallback timeout.